

A PROJECT REPORT ON

**A LINE FOLLOWING ROBOT WITHOUT USING A
MICROCONTROLLER**

SUBMITTED TO:

**ELECTRICAL AND ELECTRONICS ENGINEERING
DEPARTMENT**

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ABSTRACT

This project presents the design, implementation, and analysis of a line-following robot developed without a microcontroller. The system is based entirely on analogue and basic digital electronic components, demonstrating practical application of comparator circuits, sensor interfacing, and motor driver control techniques studied in Electronic Circuits.

The robot detects and follows a predefined black line on a white surface using infrared (IR) transmitter–receiver sensor pairs. Variations in reflected infrared light intensity are converted into voltage signals, which are processed using the LM358 operational amplifier configured as a voltage comparator. The comparator outputs generate digital control signals that drive an L293D H-bridge motor driver IC, enabling directional control of the DC motors.

The design emphasizes real-time hardware-based decision-making without software intervention. Circuit design calculations, component selection criteria, and simulation results are presented to validate system functionality. The performance of the robot is evaluated under different path conditions, and limitations such as sensitivity tuning and noise effects are analysed.

This project reinforces understanding of analogue signal processing, threshold detection, motor driver interfacing, and practical hardware implementation of autonomous control systems.

CONTENTS

- I. Introduction
- II. Objectives
- III. Components and Materials
- IV. System Overview
- V. Theory of Operation
- VI. Circuit Description
- VII. Working Principle
- VIII. Assembly Procedure
- IX. Testing and Results
- X. Conclusion
- XI. Prototype
- XII. References

INTRODUCTION

1.1 Background of Line Following Robots

A line following robot is an autonomous mobile system designed to follow a predefined path marked by a contrasting line, typically a black line on a white surface. The robot operates by continuously detecting variations in surface reflectivity and adjusting its motion accordingly to remain on track.

Line following robots are widely used in industrial automation, particularly in Automated Guided Vehicles (AGVs) for material handling and warehouse transport systems. They are also commonly employed in robotics research and education to demonstrate principles of sensing, control systems, and embedded hardware design.

The fundamental operation of a line follower involves three major stages:

1. Surface detection using sensors
2. Signal processing and decision making
3. Motor control for directional correction

In many modern implementations, these functions are executed using microcontrollers.

However, the same behaviour can be achieved using analogue and digital electronic circuits.

1.2 Motivation for Microcontroller-Free design

Most contemporary robotic systems rely on programmable microcontrollers for logic control.

While microcontrollers offer flexibility and software-based decision-making, they abstract away many fundamental electronic principles.

This project intentionally avoids the use of a microcontroller to:

- Strengthen understanding of analogue signal processing,
- Apply operational amplifier principles in comparator mode,
- Implement hardware-based logic control,
- Design motor driving circuits using H-bridge configurations.

By relying solely on hardware components such as infrared sensors, the LM358 operational amplifier, and the L293D motor driver IC, the system performs real-time decision making without software intervention.

This approach emphasizes core Electronic Circuits concepts, including voltage comparison, threshold detection, signal conditioning, and power interfacing.

1.3 Scope of the Project

The project covers the complete design process of a two-sensor line following robot, including:

- Sensor interfacing and signal conversion,
- Comparator circuit design and threshold adjustment,
- Motor driver configuration,
- Circuit simulation and verification,
- Hardware implementation and testing.

The analysis includes design calculations, system logic evaluation, and performance assessment under practical conditions.

OBJECTIVES

- I. To build a robot that can follow a predefined line automatically.
- II. To understand how infrared sensors detect surface differences.
- III. To learn how motor drivers control robot movement based on sensor feedback.
- IV. To gain hands-on experience with basic robotics components.

COMPONENTS AND MATERIALS

- I. IR LEDs, Transmitter and Receiver
- II. Resistors: 100k Ω , 220 Ω , 1k Ω
- III. LEDs
- IV. Potentiometer
- V. LM358 IC
- VI. L293D IC
- VII. BO Motors and Wheels
- VIII. Header Pins
- IX. Breadboard
- X. Wires
- XI. Transistors
- XII. Chassis
- XIII. 9V Battery (along with battery clips)

SYSTEM OVERVIEW

4.1 Block Diagram Description

The line following robot consists of four major functional blocks:

1. Sensing Stage
2. Signal Processing Stage
3. Motor Driver Stage
4. Actuation Stage

The overall system operation follows a sequential signal flow:

Surface → Sensor → Comparator → Motor Driver → Motors → Motion Correction

4.2 System Architecture and Signal Flow

The robot operates based on continuous detection of surface reflectivity.

1. Sensing Stage

The two infrared transmitter–receiver pairs are positioned at the front of the robot. The IR transmitter emits infrared radiation toward the ground surface. The amount of reflected radiation depends on the surface colour:

- White surface gives a high reflection.
- Black surface absorbs most of the rays, therefore giving back a low reflection.

The reflected signal is received by the photodiode/phototransistor, which converts light intensity into a proportional voltage signal.

2. Signal Processing Stage

The analogue voltage from each sensor is fed into the LM358 operational amplifier configured as a voltage comparator.

Each comparator compares:

- Sensor output voltage
- Reference voltage (set using a potentiometer)

If:

- Sensor voltage $>$ Reference voltage $\rightarrow$ Comparator output HIGH
- Sensor voltage $<$ Reference voltage $\rightarrow$ Comparator output LOW

This stage converts the continuously varying analogue sensor signal into a discrete digital-level control signal.

3. Motor Driver Stage

The digital outputs from the LM358 are connected to the input pins of the L293D motor driver IC. The L293D contains internal H-bridge circuits that allow:

- Forward rotation
- Reverse rotation
- Selective motor stopping

It also isolates the low-current control circuitry from the high-current motor load.

4. Actuation Stage

The DC motors receive controlled voltage from the L293D outputs. Based on the comparator logic:

- Both sensors detect white → Both motors rotate forward
- Left sensor detects black → Left motor slows/stops → Robot turns left
- Right sensor detects black → Right motor slows/stops → Robot turns right

This creates a closed-loop corrective motion system that keeps the robot aligned with the path.

4.3 Control Logic Overview

The control behaviour of the robot is determined by the digital outputs of the two comparator channels corresponding to the left and right sensors.

Let:

- L = Left comparator output
- R = Right comparator output

The motor response is determined as follows:

L	R	Left Motor	Right Motor	Motion
1	1	ON	ON	Forward
0	1	OFF	ON	Turn Left
1	0	ON	OFF	Turn Right
0	0	OFF	OFF	Stop

Theory of Operation

The operation of the Line Follower Robot is based on the principle of optical sensing and feedback control. The system relies on the interaction between infrared radiation and surface properties to distinguish between different colours. Infrared sensors operate by emitting infrared light toward the surface beneath the robot. The behaviour of this light depends on the reflectivity of the surface. Light-coloured surfaces reflect a significant portion of the incident infrared radiation, while dark surfaces absorb most of it. This difference in reflectance results in varying signal strengths at the sensor receiver. The reflected infrared light is converted into an electrical signal whose magnitude corresponds to the detected surface condition. These analogue voltage signals are then processed by the LM358 comparator, which compares them with a predefined reference voltage. The comparator converts the analogue sensor signals into discrete logic levels that the control circuitry can interpret. The logic signals are applied to the L293D motor driver IC, which controls the direction and operation of the DC motors. The driver responds by adjusting motor rotation in response to sensor feedback. This continuous sensing and correction mechanism enables the robot to adapt its movement dynamically. Fundamentally, the robot operates as a closed-loop control system, where sensor feedback directly influences motion. The robot does not follow a preprogrammed path; instead, it continuously reacts to real-time environmental inputs.

Circuit Description

The Line Follower Robot circuit is designed by integrating three major functional sections: the sensor section, the signal processing (comparator) section, and the motor driver section. Each section performs a specific task required for autonomous navigation.

The sensor section consists of infrared (IR) transmitter and receiver pairs. The IR transmitter continuously emits infrared light toward the surface beneath the robot. Depending on the surface colour, different amounts of light are reflected to the IR receiver. Light-coloured surfaces reflect a higher intensity of infrared radiation, while dark surfaces absorb most of the light. This variation in reflected light produces corresponding voltage changes at the sensor output.

The sensor outputs are fed into the LM358 operational amplifier, configured as a comparator. In this configuration, the LM358 compares the sensor voltage with a reference voltage set using a potentiometer. When the sensor detects a white surface, the reflected light is strong, resulting in a voltage level above the reference voltage. Conversely, when the sensor detects the black line, the voltage falls below the reference level. The comparator then produces a digital-like output signal representing the detected surface condition.

These comparator outputs are applied to the L293D motor driver IC, which acts as an interface between the low-power control signals and the higher-power DC motors. The L293D controls both the direction and rotation of the motors based on logic inputs received from the comparator. By selectively activating motor inputs, the driver enables forward motion, turning, or stopping actions.

Power is supplied to the circuit through a battery source, which energizes the sensors, comparator IC, motor driver, and motors. The coordinated interaction between sensing, decision-making, and actuation allows the robot to follow the line autonomously.

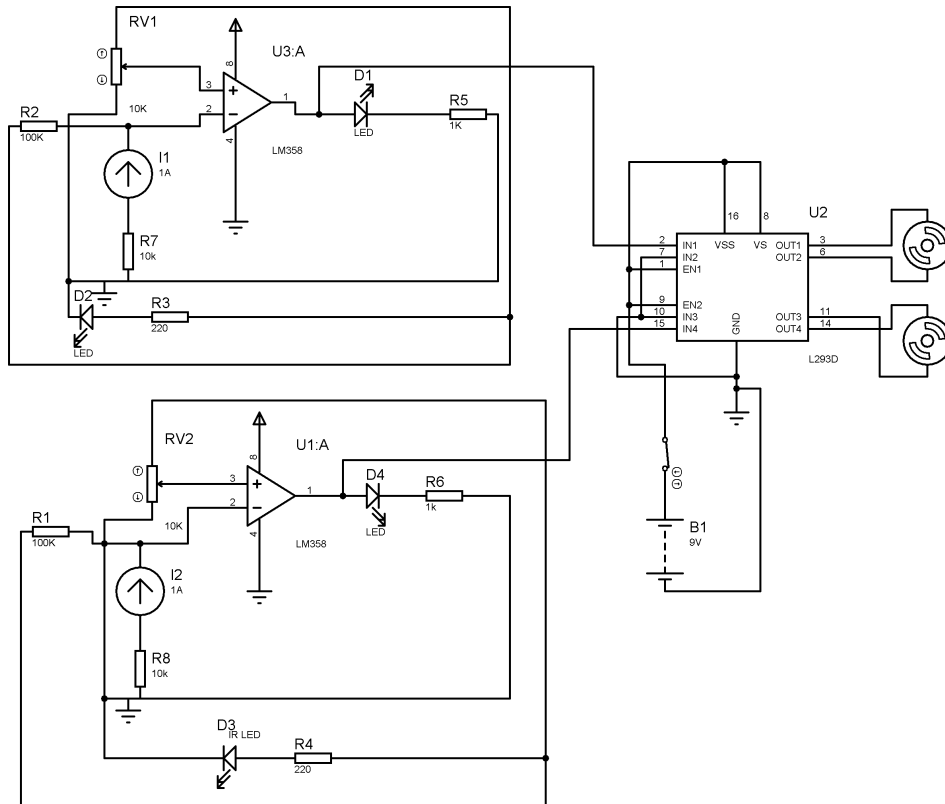


Figure 1 Circuit Diagram for the project

Working Principle

The Line Follower Robot functions by continuously monitoring the surface beneath it and making movement decisions based on sensor outputs. When the robot is placed on a track, the infrared sensors detect the presence or absence of the line. If both sensors detect the background surface (typically white), the comparator outputs maintain logic levels that command both motors to rotate forward. Under this condition, the robot moves in a straight line. When the robot deviates from the path, and one sensor detects the dark line while the other remains on the light surface, an imbalance occurs in the comparator outputs. The motor driver interprets this difference and adjusts motor activity accordingly. Typically, one motor

slows down or stops while the other continues rotating, causing the robot to turn toward the line. This corrective action continues dynamically as the robot encounters curves or directional changes. The robot's ability to follow the line is therefore achieved through continuous feedback, comparison, and motor adjustment. The system's behaviour exemplifies real-time decision-making, where movement is determined by instantaneous sensor data rather than fixed instructions. The simulation was done in Proteus. Click [here](#) to view

Assembly Procedure

The construction of the Line Follower Robot follows a systematic integration of mechanical and electrical components to ensure proper functionality:

- The assembly begins with the mechanical setup. The DC motors are securely mounted onto the robot chassis, ensuring firm alignment to avoid instability during movement. Wheels are then attached to the motor shafts to enable motion. A caster wheel is fixed at the front or rear to provide balance and smooth movement.
- Next, the infrared sensor modules are mounted at the front of the chassis. Proper positioning of the sensors is critical; they must be placed close enough to the ground for accurate detection while maintaining sufficient clearance to prevent physical contact with the surface.
- Following mechanical installation, the electronic circuit assembly is carried out on a breadboard. The LM358 comparator IC and L293D motor driver IC are mounted onto the board. Supporting components such as resistors, LEDs, and potentiometers are then placed according to the circuit design.
- Electrical connections are then established. The sensor outputs are wired to the comparator inputs. The comparator outputs are connected to the motor driver input pins. The motor driver

output pins are wired to the DC motors. Power supply connections are made to provide voltage to all circuit sections.

- Finally, the battery pack is installed and connected. All wiring is thoroughly checked to verify correct polarity and continuity. The robot is then powered on and tested. Adjustments to sensor sensitivity are made using the potentiometer to achieve stable line detection.

Testing and Results

After completing the assembly of the Line Follower Robot, systematic testing was conducted to evaluate the performance and reliability of the system. The robot was tested on a predefined track consisting of a black line drawn on a white surface, which provided sufficient contrast for accurate sensor detection. During the initial testing phase, the robot's power supply connections and motor operations were verified to ensure correct functionality. The infrared sensors were calibrated using the preset resistors to achieve optimal sensitivity. Proper calibration was necessary to allow the sensors to reliably differentiate between the line and the background under ambient lighting conditions. When placed on the test track, the sensors successfully detected variations in surface reflectivity. The LM358 comparator accurately converted the sensor outputs into logic signals, which were then processed by the L293D motor driver. The robot demonstrated forward motion when both sensors detected the background surface. As the robot encountered deviations from the line, corrective steering actions were observed. When one sensor detected the black line, the motor driver adjusted wheel rotation, enabling the robot to re-align itself with the path. The robot responded effectively to gradual curves and moderate directional changes. However, minor tracking deviations were observed during sharp turns. These deviations were primarily attributed to mechanical factors such as motor speed variations and sensor placement limitations. Despite

these minor inaccuracies, the robot maintained stable operation and successfully followed the line for most track configurations. Overall, the experimental results confirmed that the robot achieved its intended objective of autonomous line-following behaviour through sensor-based feedback control.

Conclusion

The Line Follower Robot project successfully demonstrated the practical implementation of sensor-based autonomous navigation. The system effectively integrated infrared sensing, signal processing, and motor driving mechanisms to achieve real-time path tracking.

Through this project, key engineering concepts such as optical sensing, comparator operation, motor driver functionality, and feedback control systems were explored and applied. The robot's performance validated the effectiveness of using infrared sensors to detect surface contrast and guide movement decisions. Although minor limitations were observed, particularly during sharp turns, the robot exhibited reliable and consistent behaviour under normal operating conditions. These limitations provide opportunities for further improvement, including enhanced control strategies, additional sensors, and speed regulation techniques. The project provided valuable hands-on experience in circuit design, component integration, troubleshooting, and system calibration. It also reinforced the importance of feedback mechanisms in automated systems. In conclusion, the Line Follower Robot represents a fundamental yet powerful demonstration of robotics principles and serves as a strong foundation for more advanced autonomous system designs. The Line Follower Robot can follow a line on its own, showing how sensors and motors work together to make simple robots move and respond to their environment.

✚ PICTURE OF PROTOTYPE

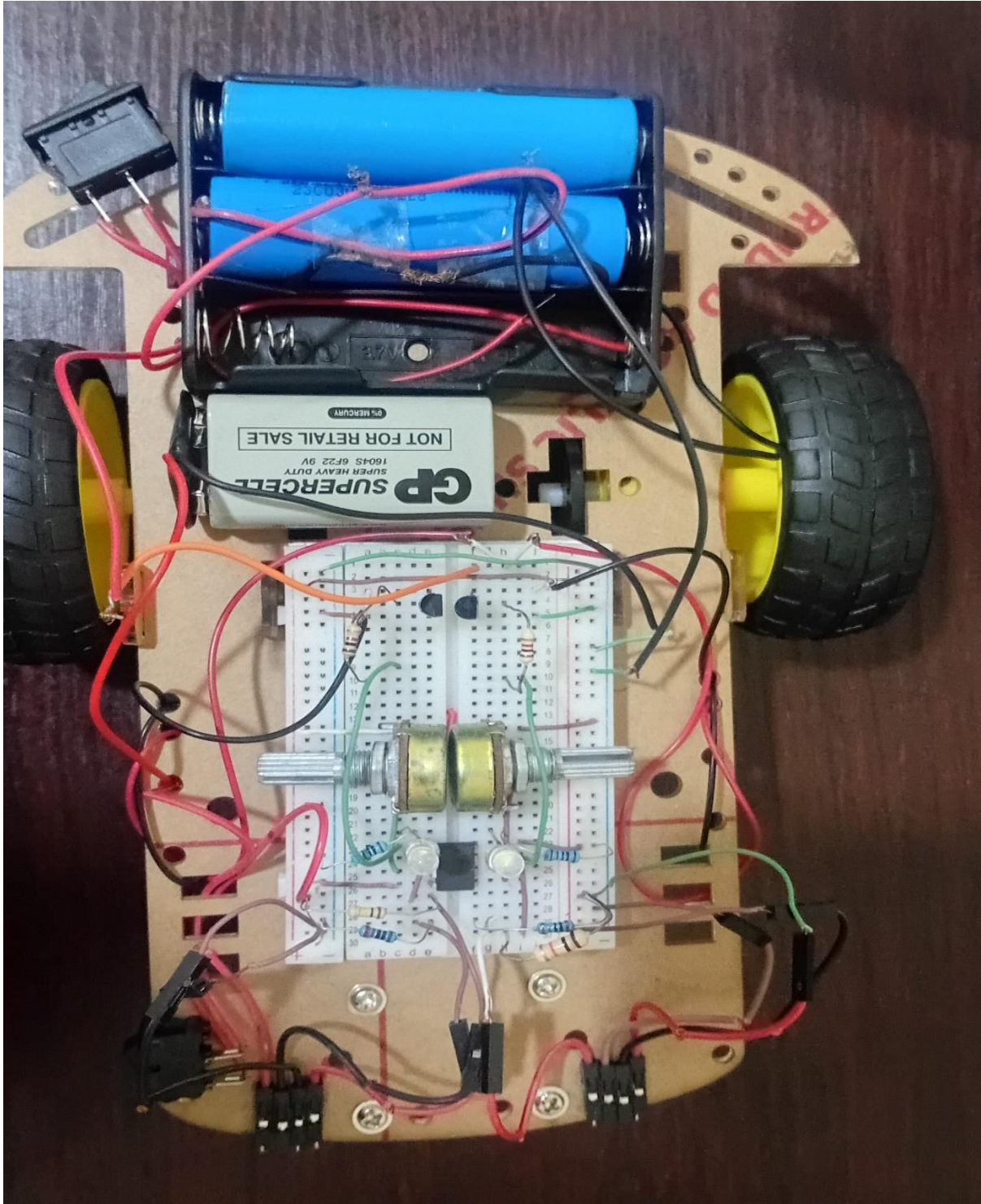


Figure 2Prototype of the project

References

- I. How to Make a Line Following Robot without a Microcontroller, [Circuit Digest](#)
- II. Course materials for Electronic Circuits I (EEG 331)
- III. [All About Circuits: Infrared Sensor Fundamentals.](#)

